

Pre-Requisites

- Conceptual understanding of neural networks — layers, activations, and weight updates.
- Basic familiarity with image data (pixels, RGB channels, resolution).
- Awareness of overfitting, model complexity, and the role of feature extraction in ML.
- No coding experience required; this is a conceptual, theory-driven session.

Session Summary

This 3-hour live session introduces learners to the **concepts and applications of Convolutional Neural Networks (CNNs)** for image analysis. Learners will understand how CNNs extract spatial features, identify visual patterns, and learn hierarchical representations from raw images. The instructor will cover the key building blocks of CNNs — convolution operations, filters, feature maps, pooling, and fully connected layers — through visual demonstrations and analogies.

Note: This is a theory-focused session. Learners are encouraged to participate actively, ask questions, and connect CNN concepts to practical examples such as image classification, object detection, and medical image analysis.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Explain the motivation behind CNNs and why traditional neural networks struggle with image data.
- Describe the key components of CNNs: convolution layers, filters, feature maps, and pooling.
- Understand how CNNs learn spatial hierarchies and local-to-global features in images.
- Interpret how convolution and pooling reduce parameters while preserving essential information.
- Explain how CNN architectures process images for classification and object recognition tasks.
- Relate CNN concepts to real-world applications in domains like healthcare, retail, and autonomous systems.

Brief Agenda

1. **Introduction & Motivation** (20 mins)
Overview of CNNs and their importance in modern deep learning. Discuss why traditional ANNs are not efficient for image data.
2. **Understanding Image Data** (25 mins)
Explain how images are represented as pixel grids, RGB channels, and tensors. Connect image structure to input for CNNs.
3. **Core Concept: Convolution Operation** (35 mins)
Introduce the idea of kernels/filters and how convolution extracts spatial features like edges and textures.
4. **Break** (10 mins)

5. Feature Maps & Pooling Layers (30 mins)

Explain feature extraction and dimensionality reduction. Introduce pooling layers and their effect on feature invariance.

6. Building a CNN Architecture (30 mins)

Discuss the role of convolutional, pooling, and fully connected layers in end-to-end CNN pipelines.

7. Understanding Hierarchical Learning (20 mins)

Show how early layers detect edges/textures and deeper layers identify complex patterns or objects.

8. Q&A & Discussion (15 mins)

Address learner questions and discuss how CNN concepts can extend to other data types.

9. Conclusion & What's Ahead (10 mins)

Summarize CNN foundations and preview next steps such as transfer learning and advanced architectures (ResNet, VGG, Inception).

Detailed Agenda

Sample Start (T+)	Duration (min)	Slot Title	Objective
00:00	20	Introduction & Motivation	Explain the challenges of processing image data using traditional ANNs. Introduce CNNs as a specialized architecture designed for spatial data.
00:20	20	Understanding Image Data	Explain how images are represented numerically — pixel grids, channels, and tensors. Discuss why locality and spatial correlation matter in image analysis.
00:40	30	Convolution Operation	Introduce convolution, filters, and feature extraction. Demonstrate visually how filters capture edges and patterns across an image.
01:10	10	Break	Short break for learners to refresh before deep-dive concepts.
01:20	20	Feature Maps & Pooling Layers	Explain the purpose of pooling layers for downsampling and translation invariance. Show how convolution and pooling work together to reduce dimensionality.
01:40	30	Building a CNN Architecture	Discuss the structure of a CNN — convolutional, pooling, flattening, and fully connected layers. Show how data flows through these stages in classification tasks.
02:10	25	Hierarchical Feature Learning	Illustrate how CNNs learn low-level to high-level features. Show example feature visualizations to build intuition on learned representations.
02:35	15	Q&A & Discussion	Discuss how CNNs are applied in industries like healthcare, automotive, e-commerce, and surveillance. Encourage learners to share related ideas.
02:50	10	Conclusion & What's Ahead	Summarize key takeaways from the session and introduce advanced CNN architectures (LeNet, AlexNet, VGG, ResNet). Provide a glimpse of transfer learning in practice.

Conclusions

By the end of this live session, learners will have developed a clear conceptual understanding of how Convolutional Neural Networks process image data. They will understand how convolution, filters, and pooling enable feature extraction and dimensionality reduction while preserving spatial information. Learners will also be able to explain the overall CNN architecture and recognize its wide-ranging applications in industry. This foundational understanding will prepare them to explore advanced CNN architectures and hands-on implementations in future sessions.